

$$e) \quad z / f(x, y) = x^3 + y^3 - 3x - 12y + 20$$

$$f_x = 3x^2 - 3$$

$$B \quad f_{xy} = 0$$

$$A \quad f_{xx} = 6x$$

$$f_y = 3y^2 - 12$$

$$f_{yx} = 0$$

$$C \quad f_{yy} = 6y$$

$$\therefore \quad D = AC - B^2$$

$$= 6x(6y) - 0^2$$

$$D = 12xy$$

$$f_x = 0$$

$$f_y = 0$$

$$3x^2 - 3 = 0$$

$$3y^2 - 12 = 0$$

$$x^2 - 1 = 0$$

$$y^2 - 4 = 0$$

$$x = \pm 1$$

$$y = \pm 2$$

$$D = 12xy$$

$$A = 6x$$

$$B = 0$$

$$C = 6y$$

$$D$$

$$(1, 2)$$

$$6$$

$$0$$

$$12$$

$$72$$

min

$$(-1, -2)$$

$$-6$$

$$0$$

$$-12$$

$$72$$

max

$$(-1, 2)$$

$$-6$$

$$0$$

$$12$$

$$-72$$

saddle

$$(1, -2)$$

$$6$$

$$0$$

$$-12$$

$$-72$$

saddle

$$b) \quad f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$$

$$f_x = 4x^3 - 4x + 4y$$

$$f_y = 4y^3 + 4x - 4y$$

$$A \quad f_{xx} = 12x^2 - 4$$

$$C \quad f_{yy} = 12y^2 - 4$$

$$C \quad f_{xy} = 0$$

$$f_x = 0$$

$$f_y = 0$$

$$4x^3 - 4x + 4y = 0$$

$$4y^3 + 4x - 4y = 0$$

$$4x = 4y - 4y^3$$

$$x = y - y^3$$

$$\begin{aligned} 4x^3 - 4x + 4y \\ + 4y^3 + 4x - 4y \end{aligned}$$

$$4x^3 + 4y^3 = 0$$

$$4x^3 = -4y^3$$

$$\sqrt[3]{x^3} = \sqrt[3]{-y^3}$$

$$x = -y$$

$$4x^3 - 4x + 4y = 0$$

$$4(-y)^3 - 4(-y) + 4y = 0$$

$$-4y^3 + 4y + 4y = 0$$

$$-4y^3 + 8y = 0$$

$$y(-4y^2 + 8) = 0$$

$$y = 0$$

$$-4y^2 + 8 = 0$$

$$y^2 = \frac{8}{4}$$

$$y = \pm\sqrt{2}$$

$$0$$

$$+\sqrt{2}$$

$$-\sqrt{2}$$

$$e) f(x, y) = x^2 - xy + y^2 - 2x + y$$

$$f_x = 2x - y - 2$$

$$f_y = -x + 2y + 1$$

$$A \quad f_{xx} = 2$$

$$C \quad f_{yy} = 2$$

$$B \quad f_{xy} = 0$$

$$f_x = 0$$

$$f_y = 0$$

$$2x - y - 2 = 0$$

$$-x + 2y + 1 = 0$$

$$2x - 2 = y$$

$$-x + 2(2x - 2) + 1 = 0$$

$$-x + 4x - 4 + 1 = 0$$

$$y = 2 - 2$$

$$3x - 3 = 0$$

$$y = 0$$

$$3x = 3$$

$$x = 1$$

$$(1, 0)$$

$$e) f(x, y) = x^3 y^2 (1 - x - y)$$

$$= x^3 y^2 - x^4 y^2 - x^3 y^3$$

$$f_x = 3x^2 y^2 - 4x^3 y^2 - 3x^2 y^3$$

$$f_y = 2x^3 y - 2x^4 y - 3x^3 y^2$$

$$A \quad f_{xx} = 6xy^2 - 12x^2 y^2 - 6xy^3$$

$$C \quad f_{yy} = 2x^3 - 2x^4 - 6x^3 y$$

$$B \quad f_{xy} = 12xy - 24x^2 y - 18x y^2$$

